



Module 2- Transformers

Ragi R. Menon

Assistant Professor

Division of Electrical Engineering

SOE

Introduction

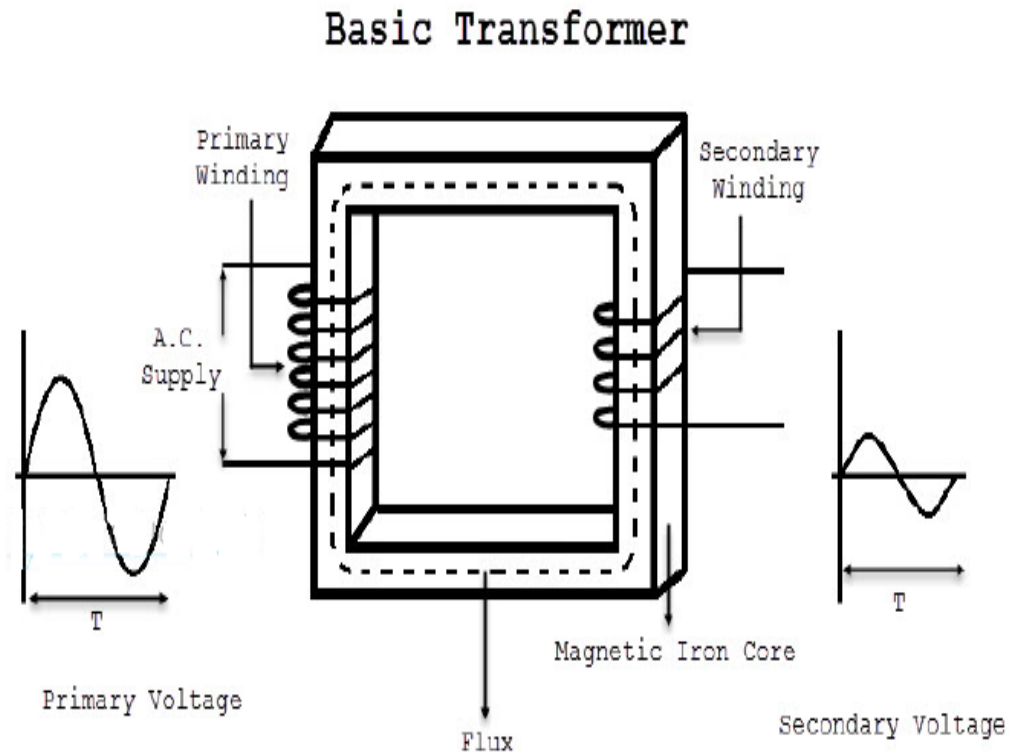
Why we need transformers?

- Electrical energy is generated at places where it is easier to get waterhead, oil, coal etc. Then energy is to be transmitted at considerable distances for use in villages, town etc. So a device is required to step up voltage at generating station
- Stepping down the same at the places where it is used also require some device.
- In our country, electrical energy is usually generated at 11 or 33kV, in order to overcome losses it is stepped up to 220kV, 400kV, 765kV with the help of step up transformer for transmission and then stepped down to 66kV or 33kV at grid substations for feeding various substations which further stepped down to 11kV for feeding distribution transformers stepping down further to 230V for the consumer uses with the help of step down transformer.

Transformer

Transformer is a static AC machine that (i) transfers electric power from one circuit to another circuit (ii) does so without a change of frequency (iii) does so by the principle of Electromagnetic Induction (iv) has electric circuits that are commonly linked by magnetic circuit (No direct electrical contact)

Working Principle:- Mutual Induction



- A transformer consists of two coils with high mutual inductance that are electrically separated but have common magnetic circuit.
 - The first set of the coil, which is called as the Primary Coil or Primary Winding, is connected to an alternating voltage source called Primary Voltage.
 - The other coil, which is called as Secondary Coil or Secondary Winding, is connected to the load and the load draws current depends on the resulting alternating voltage (stepped up or stepped down voltage).
 - The alternating voltage at the input excites the Primary Winding, an alternating current circulates the winding. The alternating current will result in an alternating magnetic flux, which passes through the iron magnetic core and completes its path.
 - Since the secondary winding is also linked to the alternating magnetic flux, according to Faraday's Law, an E.M.F is induced in the secondary winding. The strength of the voltage at the secondary winding is dependent on the number of secondary windings.
- $$E \propto N \frac{d\phi}{dt}$$
- Thus, without making an electrical contact, the alternating power in the primary circuit is transferred to the secondary circuit.

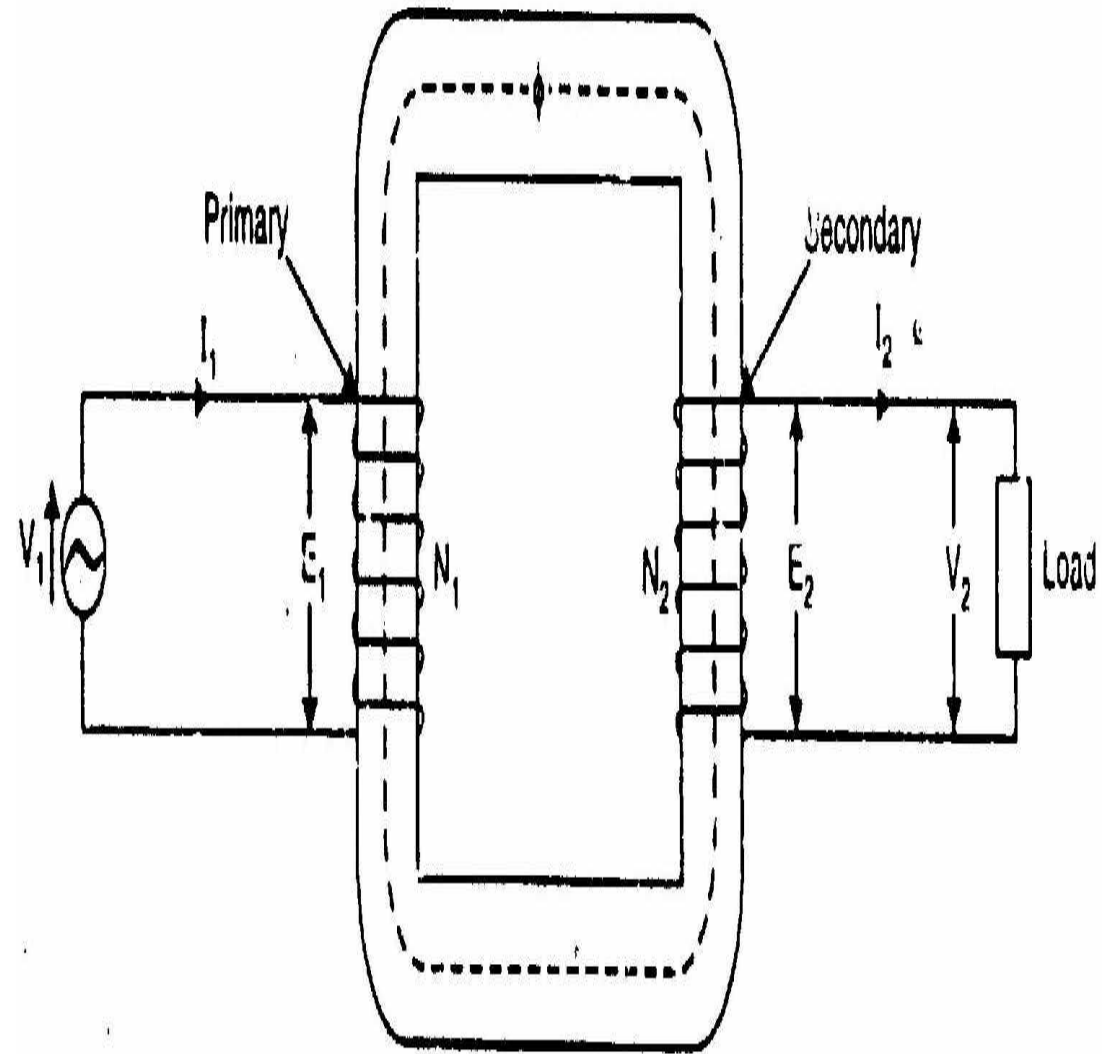
- When an alternating voltage V_1 is applied to the primary, an alternating flux ϕ is set up in the core. This alternating flux links both the windings and induces e.m.f.s E_1 and E_2 in them according to Faraday's laws of electromagnetic induction. The e.m.f. E_1 is termed as primary e.m.f. and e.m.f. E_2 is termed as secondary e.m.f.

$$E_1 = -N_1 \frac{d\phi}{dt}$$

$$E_2 = -N_2 \frac{d\phi}{dt}$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

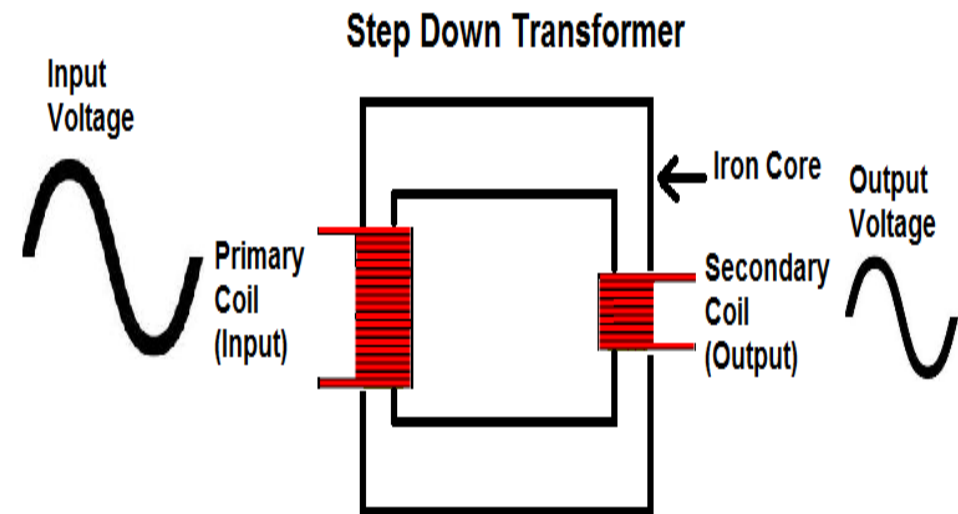
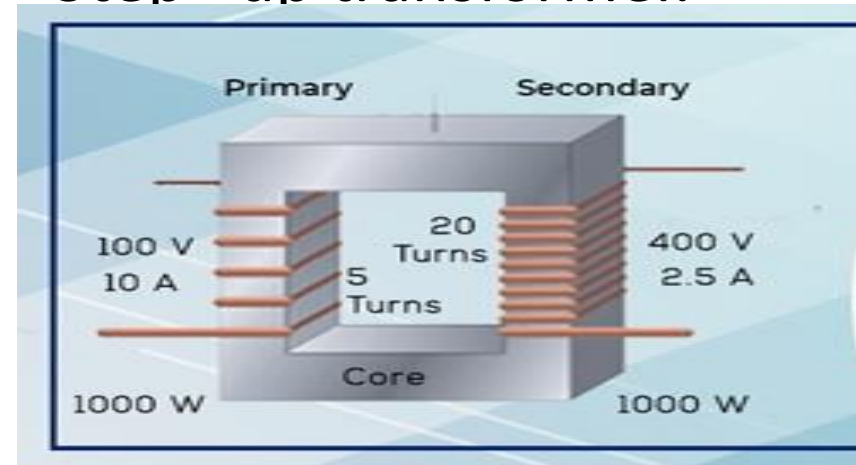
- If $N_2 > N_1$, then $E_2 > E_1$ (or $V_2 > V_1$) and we get a step-up transformer.
- On the other hand, if $N_2 < N_1$, then $E_2 < E_1$ (or $V_2 < V_1$) and we get a step-down transformer



Types of transformer

- Based on function:
 - Step - up transformer.
 - Step - down transformer.
- Based on design:
 - Core type.
 - Shell type.
- Based on application:
 - Power transformer.
 - Distribution transformer.
 - Current transformer & Potential transformer-measuring purposes.

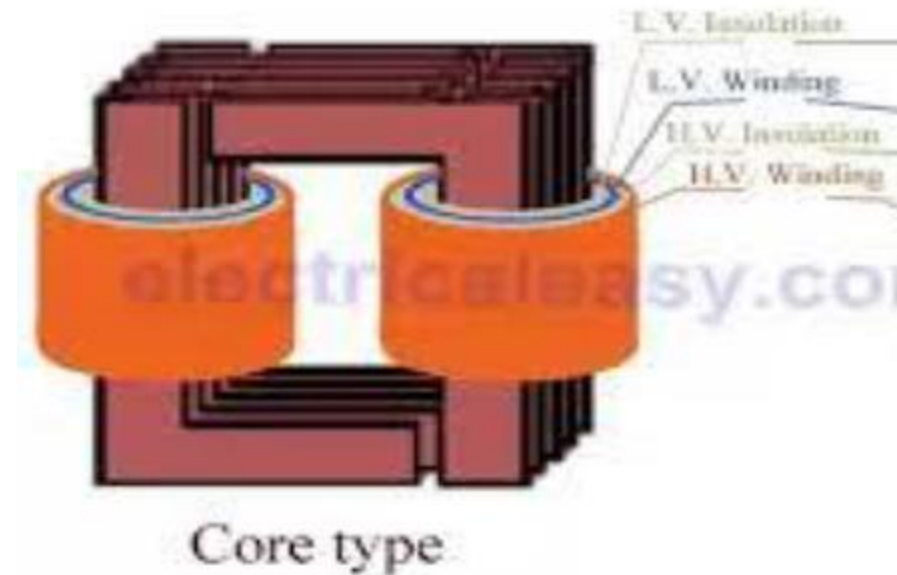
- Based on function:
 - Step - up transformer.



Based on design

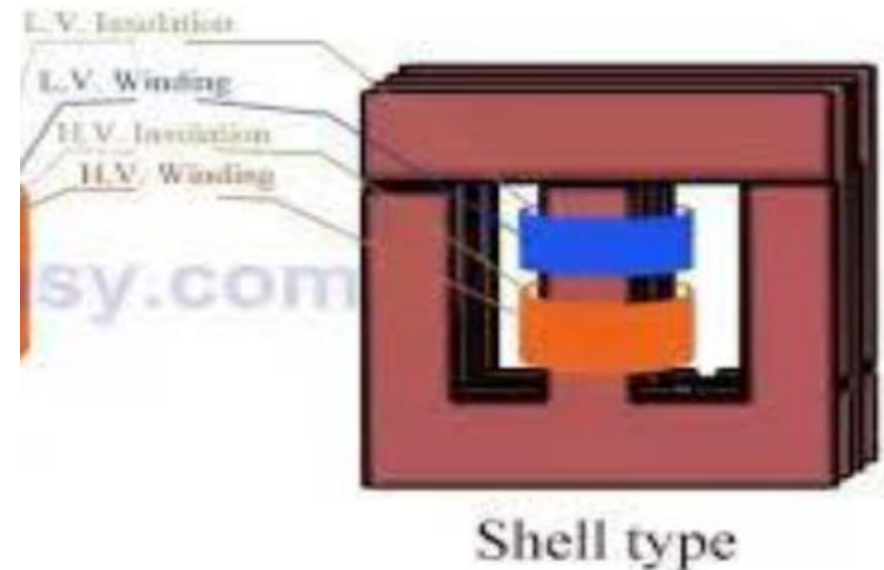
Core type transformer

- In core-type transformer, the windings covers considerable part of the core.
- The core is laminated to reduce eddy current power loss.
- Half of the primary winding and half of the secondary winding are placed on each leg.
- Insulations are provided between windings and between core & winding.
- To reduce insulation requirements, LV winding is placed near the core.



Based on design

- Shell type transformers
- Core covers the considerable portion of the windings.
- Both HV & LV windings placed around the central limb, which has twice the cross-sectional area of the two outer limbs..
- The coils are form-wound but are multi layer disc type usually wound in the form of pancakes.
- Paper is used to insulate the different layers of the multi-layer discs.



Based on application

Power Transformers

- Power transformers are used in transmission network of higher voltages for step-up and step down application (400 kV, 200 kV, 110 kV, 66 kV, 33kV) and are generally rated above 200MVA.
- Power Transformers are mainly focusing on the high voltage since it requires high insulation levels
- Power transformers does not require to be in on condition when there is no demand of power. Since ultimately loss of the power transformer is very less.
- The power transformers are designed to operate 99% of efficiency

Distribution Transformers

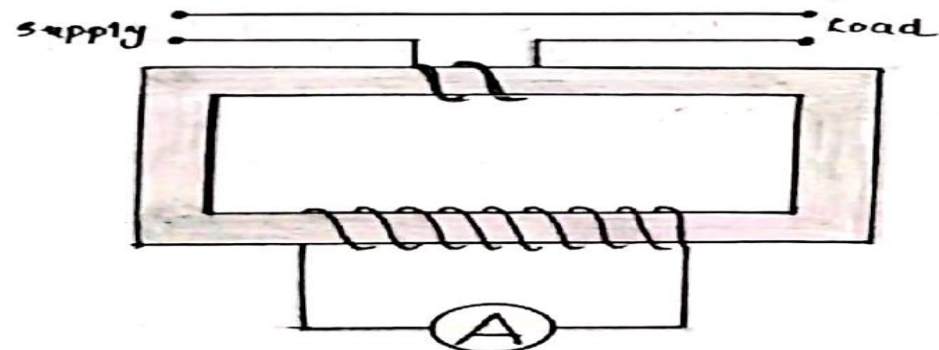
- Distribution transformer are mainly used for providing electricity to the consumer level, which means it reduces the voltage level to the consumer rating i.e 11kV/440 volts, 110 volts etc .
- In case distribution transformers are working under consumer level voltages, therefore it does not require as much insulation (less voltage level)
- In distribution transformers are working 24 hours in a day, since their losses should be high. Therefore, distribution transformers are designed for less iron losses.
- distribution transformers operates at 60 % to 75 % efficiency.

INSTRUMENT TRANSFORMERS

- * Instrument transformers are used to measure high alternating current and voltages.
- * It uses standard low-range or instruments in conjunction with instrument transformers of accurate ratio.
- * Two types:
 1. Current Transformers (CT)
 2. Potential Transformers (PT)

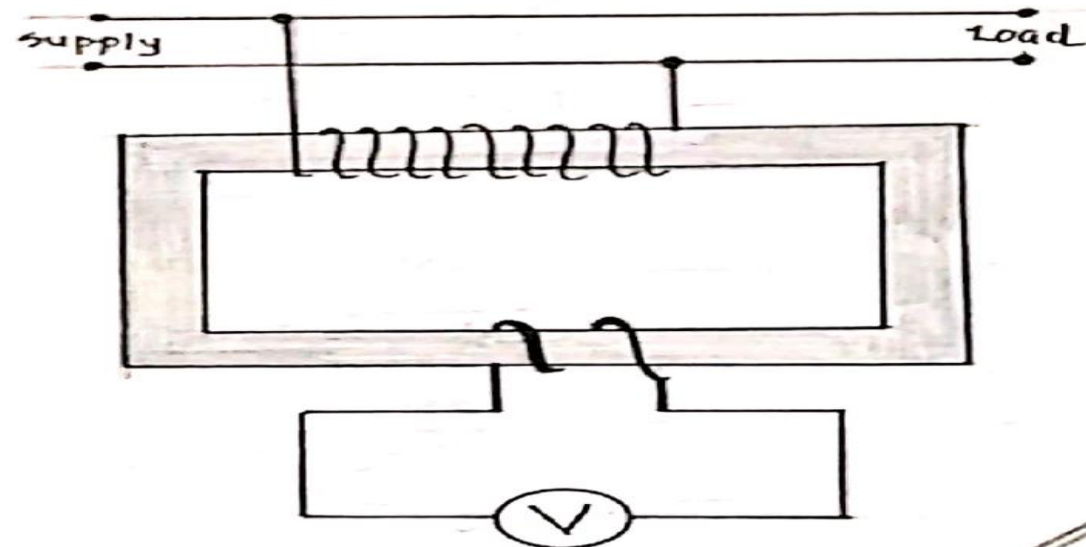
1) CURRENT TRANSFORMERS: (CT)

1. CTs are used with low-range ammeters to measure large alternating currents.
2. CTs insulate the instrument from the high voltage line but also step down the current in a known ratio.
3. CTs have a primary winding with few no: of thick turns connected in series with the line whose current is to be measured.
4. The secondary winding has large no: of fine turns connected across the ammeter terminals.
5. By knowing the current ratio (I_1/I_2) & the ammeter reading, the actual line current can be calculated.



2) POTENTIAL TRANSFORMERS: (PT)

1. PTs are used with low-range voltmeters to measure large alternating voltages.
2. PTs step down the voltage in a known ratio.
3. CTs have a primary winding with large no: of fine turns connected in parallel with the line whose voltage is to be measured.
4. The secondary windings have few no: of thick turns connected across the voltmeter terminals.
5. By knowing the voltage ratio (V_1/V_2) & the voltmeter reading, the actual voltage can be calculated.
6. Grounding is provided to ensure operator safety.



4.6.2. E.M.F. Equation of a Transformer

Consider an alternating voltage V_1 of frequency f applied to the primary of the transformer. This develops a primary current which sets up an alternating flux ϕ which can be represented as $\phi = \phi_m \sin \omega t$

The instantaneous e.m.f. induced in the primary is

$$\begin{aligned} e_1 &= -N_1 \frac{d\phi}{dt} = -N_1 \frac{d(\phi_m \sin \omega t)}{dt} \\ &= -\omega N_1 \phi_m \cos(\omega t) \\ &= -2\pi f N_1 \phi_m \cos(\omega t) \\ &= 2\pi f N_1 \phi_m \sin(\omega t - 90^\circ) \dots \dots \dots (1) \end{aligned}$$

It is clear from the above equation that maximum value of induced e.m.f. in the primary is

$$E_{m1} = 2\pi f N_1 \phi_m$$

The r.m.s. value E_1 of the primary e.m.f. is

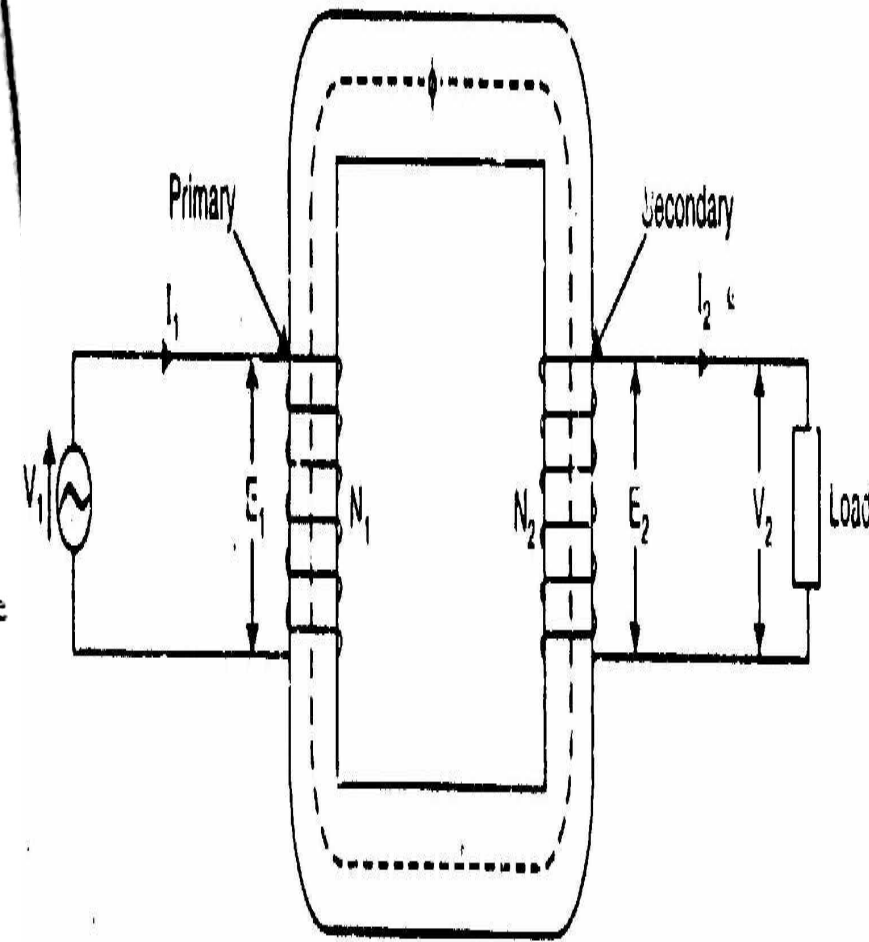
$$E_1 = \frac{E_{m1}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_m}{\sqrt{2}}$$

$$\text{or } E_1 = 4.44 f N_1 \phi_m \text{ volts}$$

since the flux ϕ is common to both windings,

$$E_2 = 4.44 f N_2 \phi_m \text{ Volts}$$

In an ideal transformer $E_1 = V_1$ and $E_2 = V_2$

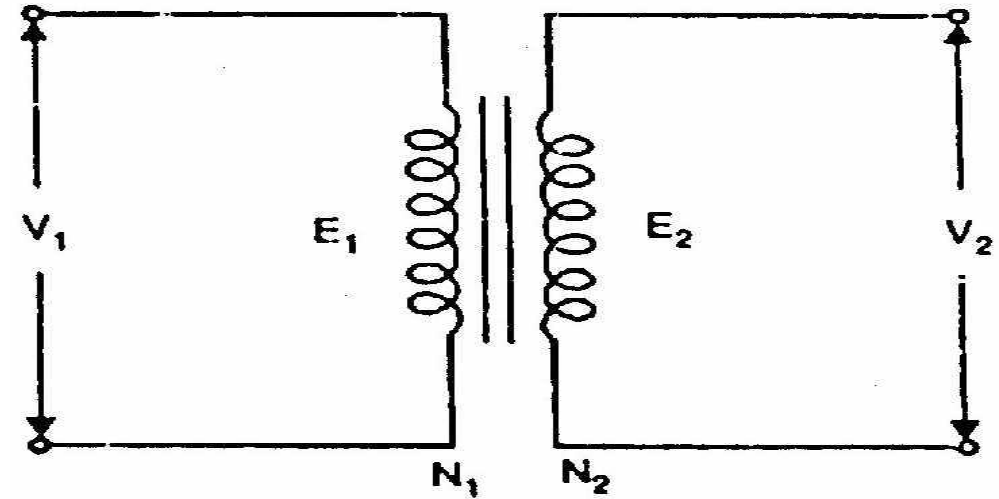


Transformation ratio

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

The constant K is called *voltage transformation ratio*

Turns ratio is inverse of transformation ratio.

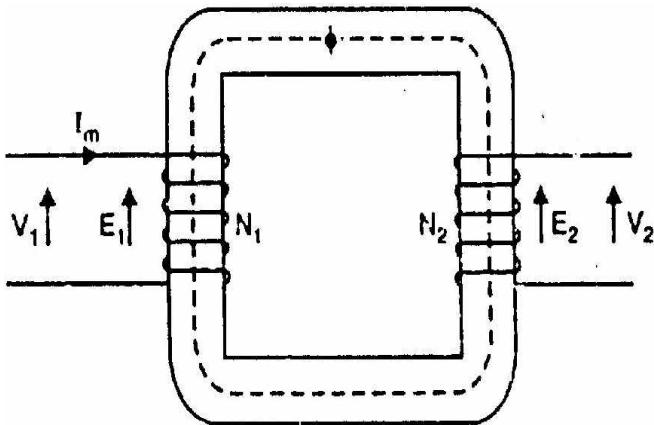


Ideal transformer

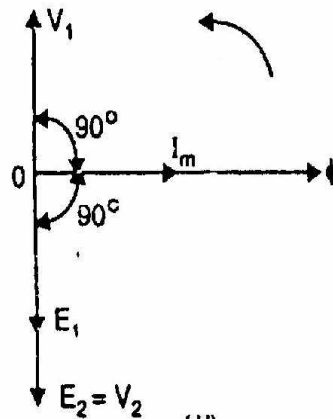
An ideal transformer is one that has

- (i) no winding resistance
- (ii) no leakage flux i.e., the same flux links both the windings
- (iii) no iron losses (i.e., eddy current and hysteresis losses) in the core

Ideal Transformer on No load



(i)



(ii)

- When an alternating voltage V_1 is applied to the primary, it draws a small magnetizing current I_m which lags behind the applied voltage by 90° . This alternating current I_m produces an alternating flux ϕ which is proportional to and in phase with it. The alternating flux ϕ links both the windings and induces e.m.f. E_1 in the primary and e.m.f. E_2 in the secondary.
- Flux ϕ is common to both the windings, it has been taken as the reference phasor
- The primary e.m.f. E_1 is, at every instant, equal to and in opposition to V_1 (Lenz's law). Both e.m.f.s E_1 and E_2 lag behind flux ϕ by 90° . However, their magnitudes depend upon the number of primary and secondary turns. Note that E_1 and E_2 are in phase. But E_1 is equal to V_1 and 180° out of phase with it

Ideal Transformer on Load

- (i) $E_1 = V_1$ and $E_2 = V_2$ as there is no voltage drop in the windings.

$$\therefore \frac{E_2}{E_1} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = K$$

- (ii) there are no losses. Therefore, volt-amperes input to the primary are equal to the output volt-amperes i.e.

$$V_1 I_1 = V_2 I_2$$

$$\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

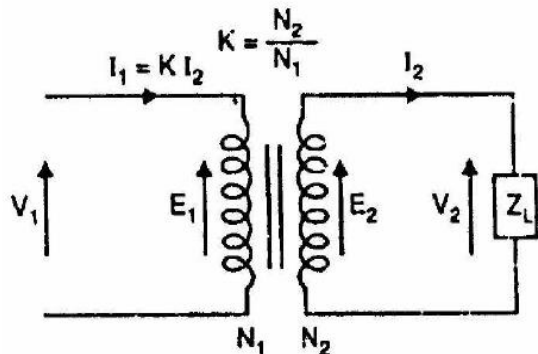
Currents are in the inverse ratio of voltage transformation ratio.

This simply means that if we raise the voltage, there is a corresponding decrease of current.

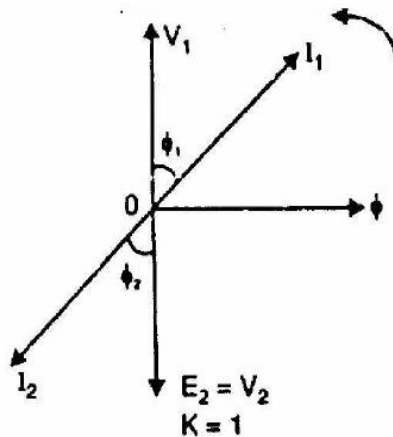
- Connect a load Z_L across the secondary of an ideal transformer. The secondary e.m.f. E_2 will cause a current I_2 to flow through the load.

$$I_2 = \frac{E_2}{Z_L} = \frac{V_2}{Z_L}$$

- The angle at which I_2 leads or lags V_2 (or E_2) depends upon the resistance and reactance of the load.



(i)



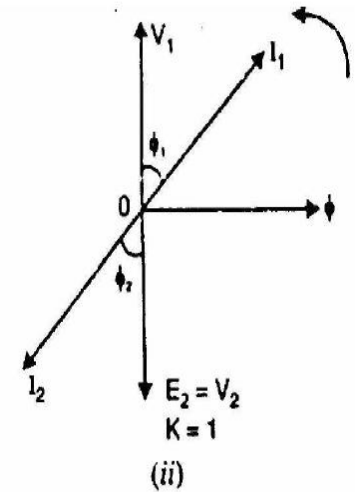
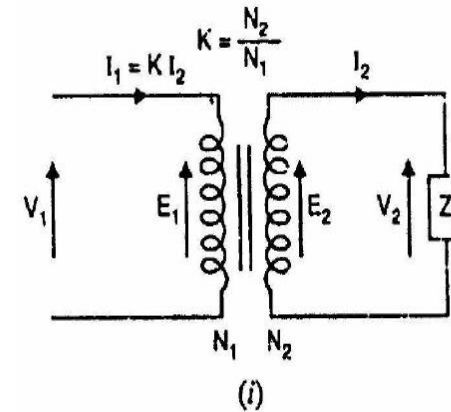
(ii)

- Here, we have considered inductive load so that current I_2 lags behind V_2 (or E_2) by ϕ_2 .
- The secondary current I_2 sets up an m.m.f. $N_2 I_2$ which produces a flux in the opposite direction to the flux ϕ originally set up in the primary by the magnetizing current. This will change the flux in the core from the original value. However, the flux in the core should not change from the original value. In order to fulfill this condition, the primary must develop an m.m.f. which exactly counterbalances the secondary m.m.f. $N_2 I_2$. Hence a primary current I_1 must flow such that

$$N_1 I_1 = N_2 I_2$$

$$I_1 = \frac{N_2}{N_1} I_2 = K I_2$$

when a transformer is loaded and carries a secondary current I_2 , then a current I_1 , ($= K I_2$) must flow in the primary to maintain the m.m.f. balance. In other words, the primary must draw enough current to neutralize the demagnetizing effect of secondary current so that mutual flux ϕ remains constant.



In drawing the phasor diagram, the value of K has been assumed unity so that primary phasors are equal to secondary phasors. The secondary current I_2 lags behind V_2 (or E_2) by ϕ_2 . It causes a primary current $I_1 = K I_2 = 1 \times I_2$ which is in antiphase with it.

(i) $\phi_1 = \phi_2$

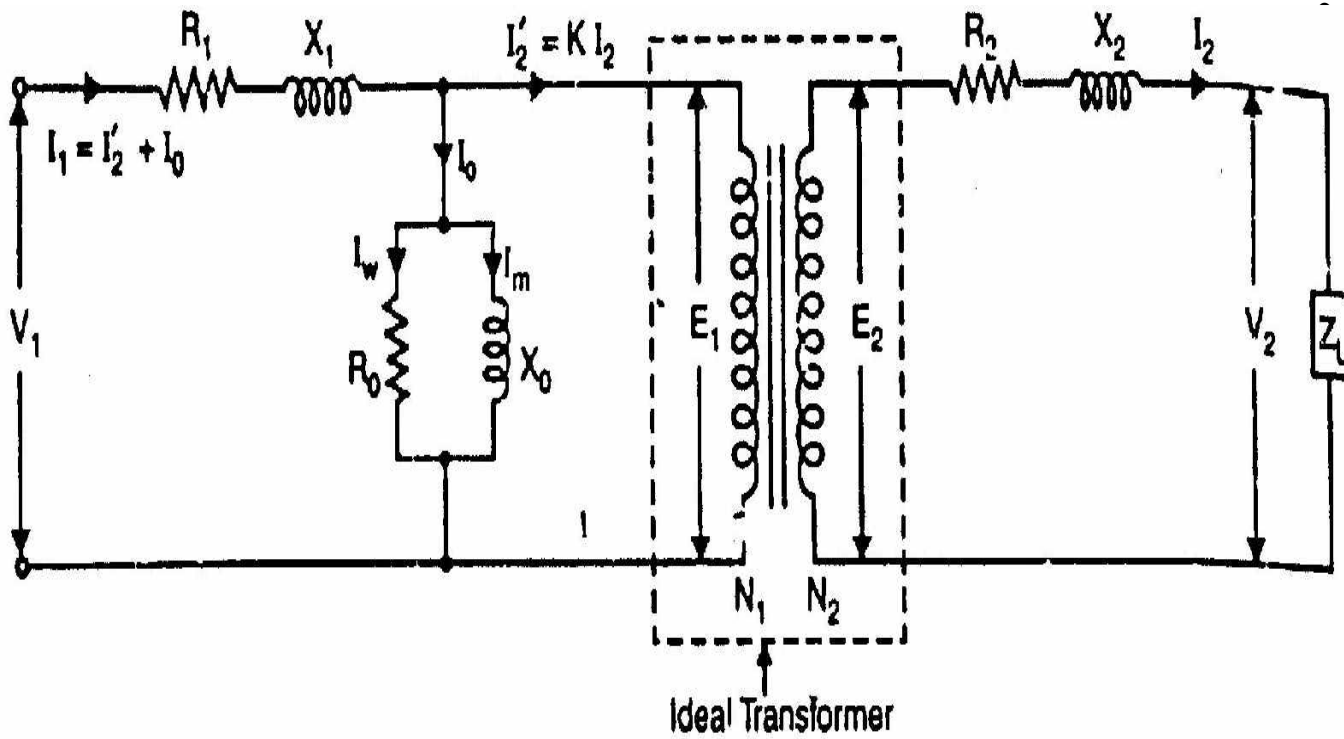
or $\cos \phi_1 = \cos \phi_2$

Thus, power factor on the primary side is equal to the power factor on the secondary side.

- (ii) Since there are no losses in an ideal transformer, input primary power is equal to the secondary output power i.e.,

$$V_1 I_1 \cos \phi_1 = V_2 I_2 \cos \phi_2$$

Equivalent circuit of a loaded practical transformer



- R_1 - primary winding resistance

- R_2 - secondary winding resistance.

X_1 - leakage reactance of primary winding

X_2 - leakage reactance of the secondary winding.

The parallel circuit $R_0 - X_0$ is the no-load

equivalent circuit of the transformer.

R_0 - core losses(hysteresis and eddy current losses) so that current I_w which supplies the core losses is shown passing through R_0 .

The inductive reactance X_0 - loss-free coil which passes the magnetizing current I_m .

Equivalent circuit

- (i) When the transformer is on no-load (i.e., secondary terminals are open circuited), there is no current in the secondary winding. However, the primary draws a small no-load current I_0 . The no-load primary current I_0 is composed of (a) magnetizing current (I_m) to create magnetic flux in the core and (b) the current I_w required to supply the core losses.
- (ii) When the secondary circuit of a transformer is closed through some external load Z_L , the voltage E_2 induced in the secondary by mutual flux will produce a secondary current I_2 . There will be $I_2 R_2$ and $I_2 X_2$ drops in the secondary winding so that load voltage V_2 will be less than E_2 .
- (iii) When the transformer is loaded to carry the secondary current I_2 , the primary current consists of two components:
 - (a) The no-load current I_0 to provide magnetizing current and the current required to supply the core losses.
 - (b) The primary current $I'_2 (= K I_2)$ required to supply the load connected to the secondary.
- (iv) Since the transformer is now ideal, the primary induced voltage E_1 can be calculated from the relation:

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

If we add $I_1 R_1$ and $I_1 X_1$ drops to E_1 , we get the primary input voltage V_1

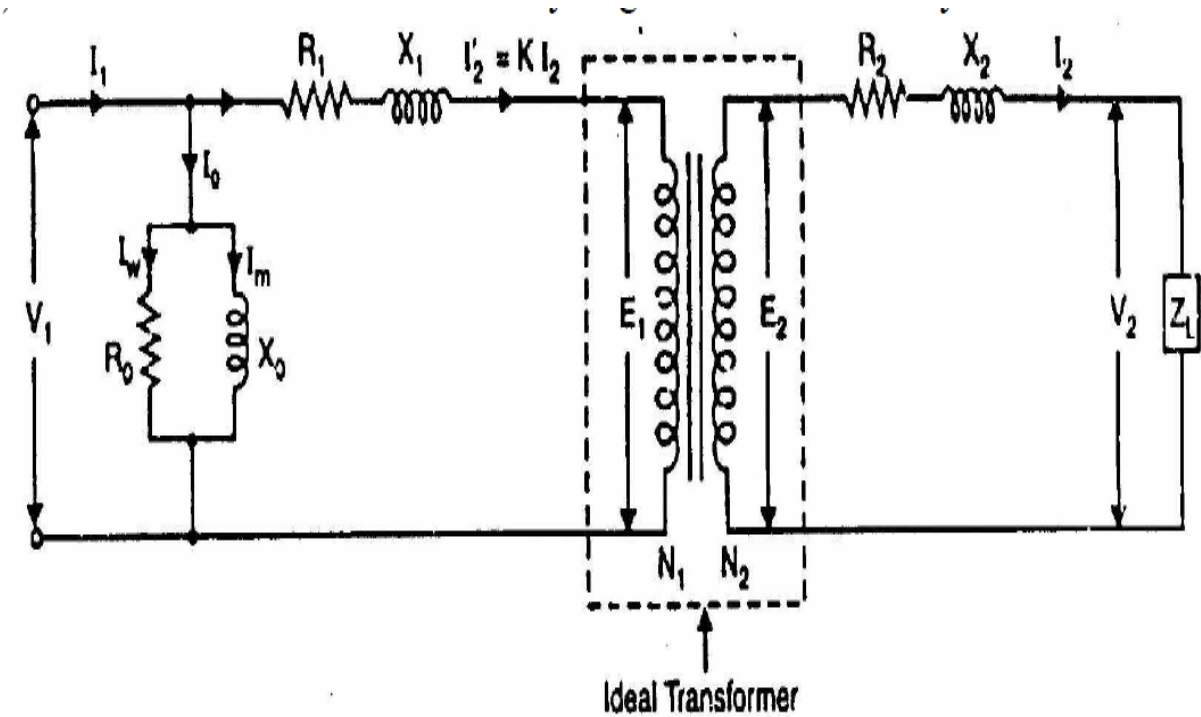
$$V_1 = E_1 + I_1 R_1 + j I_1 X_1$$

$$V_1 = E_1 + I_1 Z_1$$

$$V_2 = E_2 - I_2 (R_2 + j X_2) = E_2 - I_2 Z_2$$

Simplified Equivalent Circuit of transformer

- The no-load current I_0 of a transformer is small as compared to the rated primary current. Therefore, voltage drops in R_1 and X_1 due to I_0 are negligible.
- The equivalent circuit can be simplified by transferring the shunt circuit $R_0 - X_0$ to the input terminals as shown in Fig. This modification leads to only slight loss of accuracy.



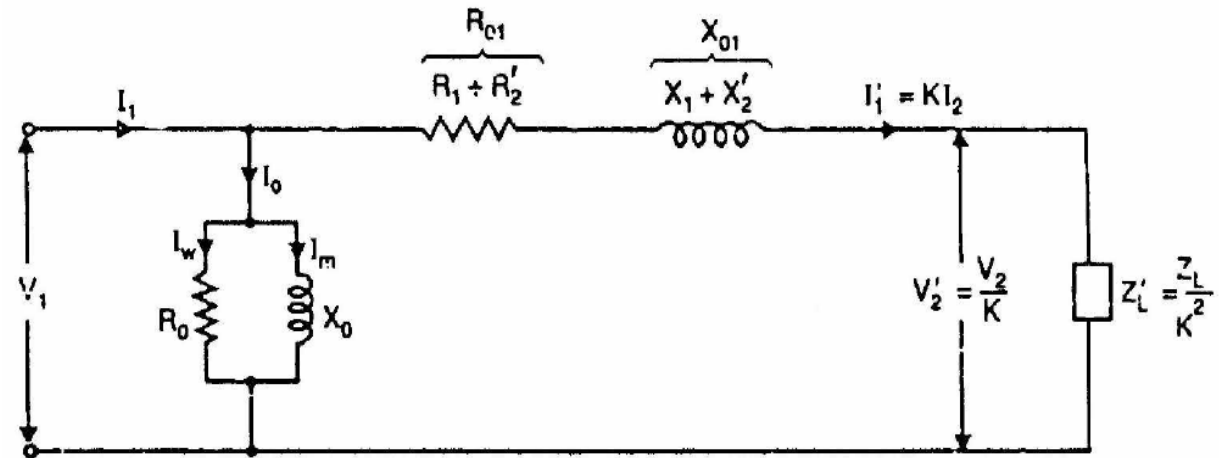
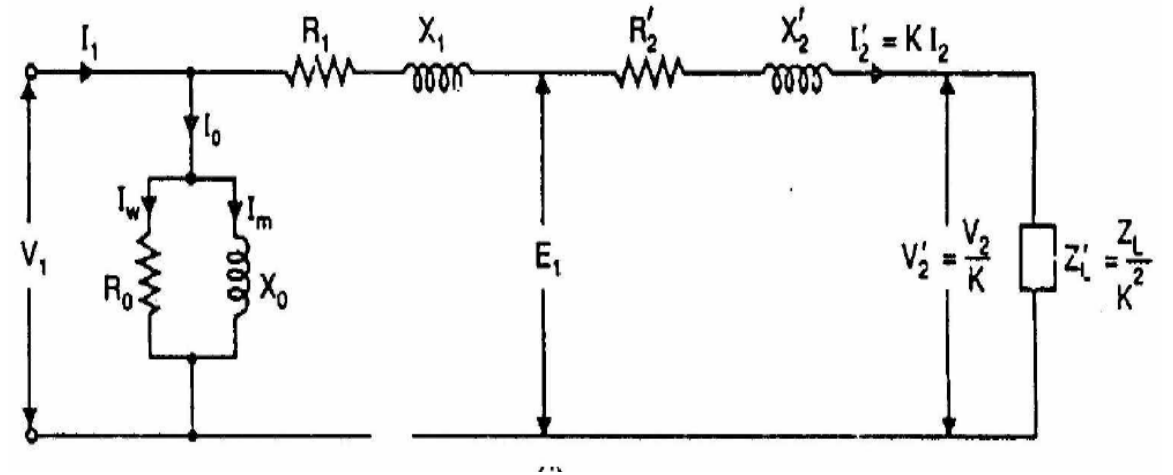
Equivalent circuit referred to primary

- If all the secondary quantities are referred to the primary, we get the equivalent circuit of the transformer referred to the primary
- Note that when secondary quantities are referred to primary, resistances/reactances/impedances are divided by K^2 ,
- voltages are divided by K and currents are multiplied by K .

$$K'_2 = \frac{R_2}{K^2}; \quad X'_2 = \frac{X_2}{K^2}; \quad Z'_L = \frac{Z_L}{K^2}; \quad V'_2 = \frac{V_2}{K}; \quad I'_2 = K I_2$$

$$Z_{01} = R_{01} + j X_{01}$$

$$R_{01} = R_1 + R'_2; \quad X_{01} = X_1 + X'_2$$



Equivalent circuit referred to secondary

- If all the primary quantities are referred to secondary, we get the equivalent circuit of the transformer referred to secondary as shown in Fig.
- This further reduces to fig. Note that when primary quantities are referred to secondary resistances/reactances/impedances are multiplied by K^2 , voltages are multiplied by K , and currents are divided by K .

Equivalent resistance of transformer referred to secondary

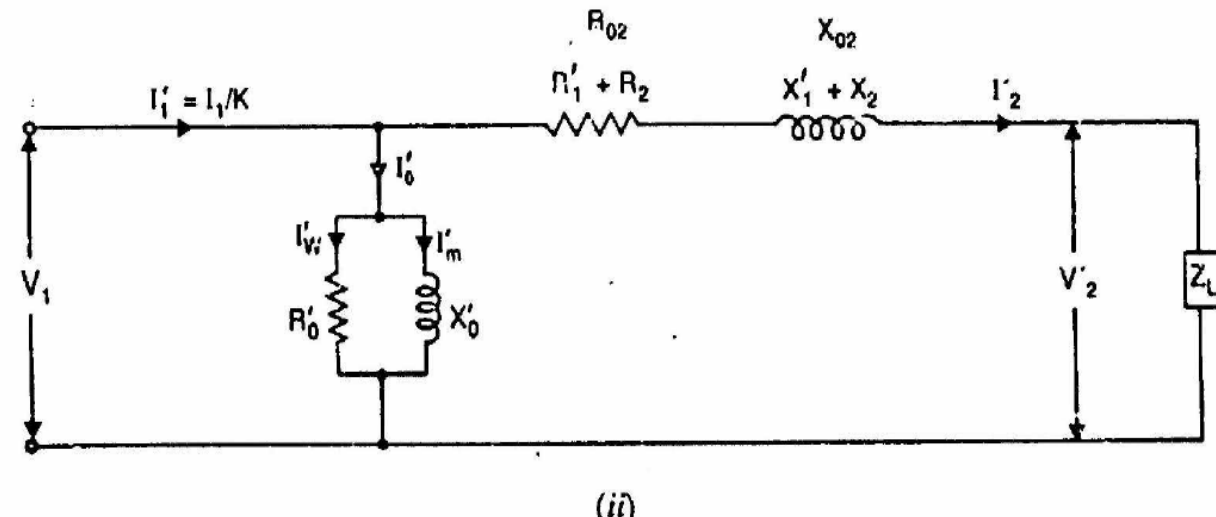
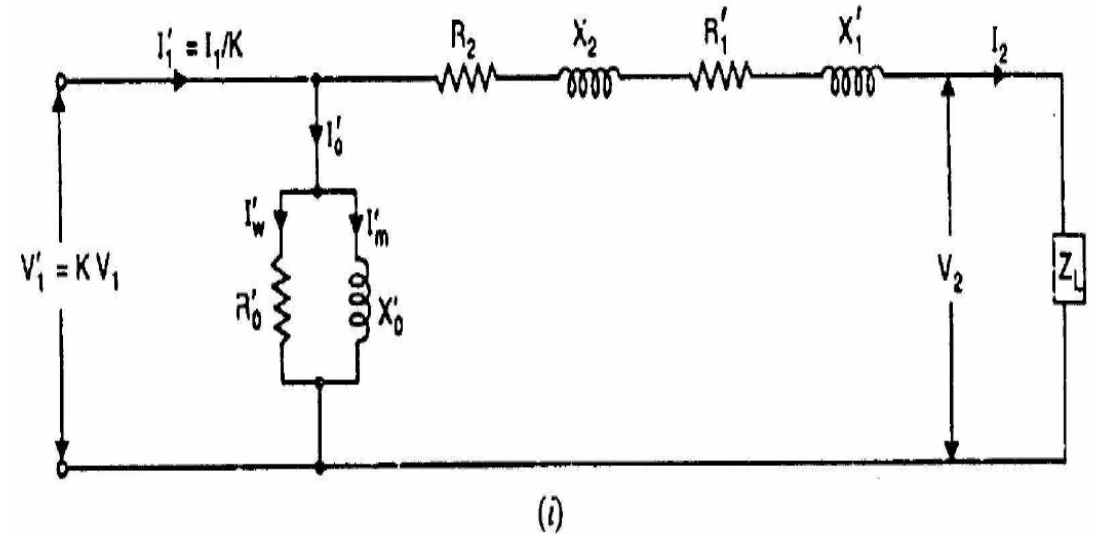
$$R_{02} = R_2 + R'_1 = R_2 + K^2 R_1$$

Equivalent reactance of transformer referred to secondary

$$X_{02} = X_2 + X'_1 = X_2 + K^2 X_1$$

Equivalent impedance of transformer referred to secondary

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$



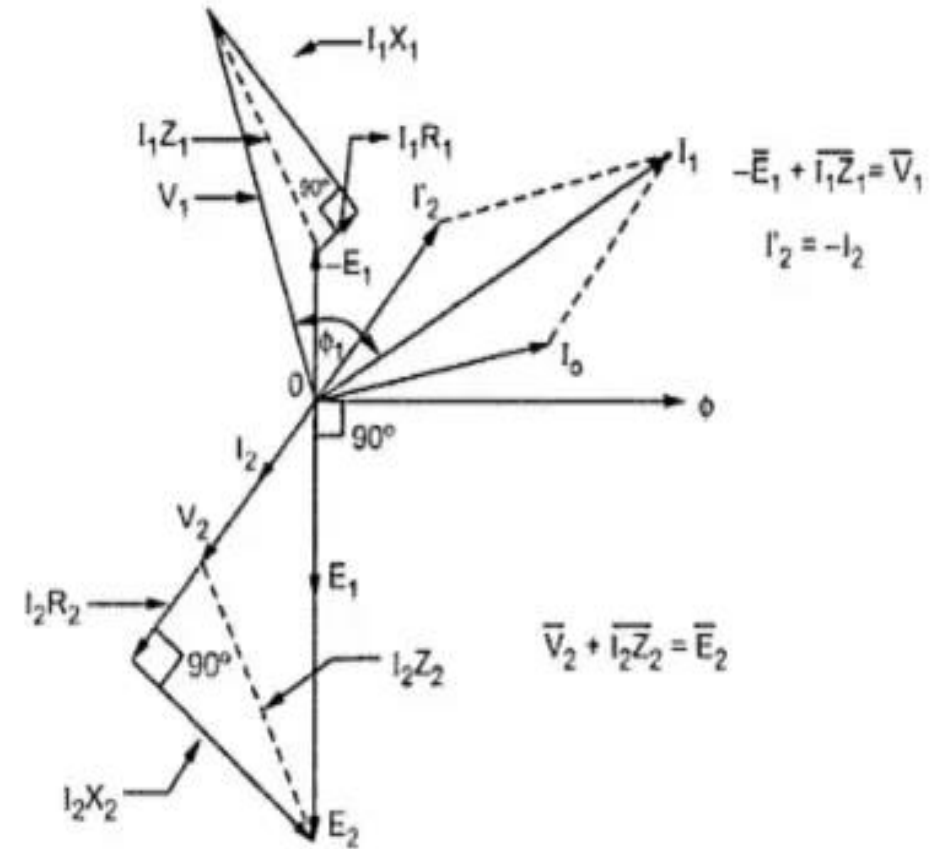
Vector Diagram

- Resistive Load (Unity Pf)

$$V_1 = E_1 + I_1 R_1 + j I_1 X_1$$

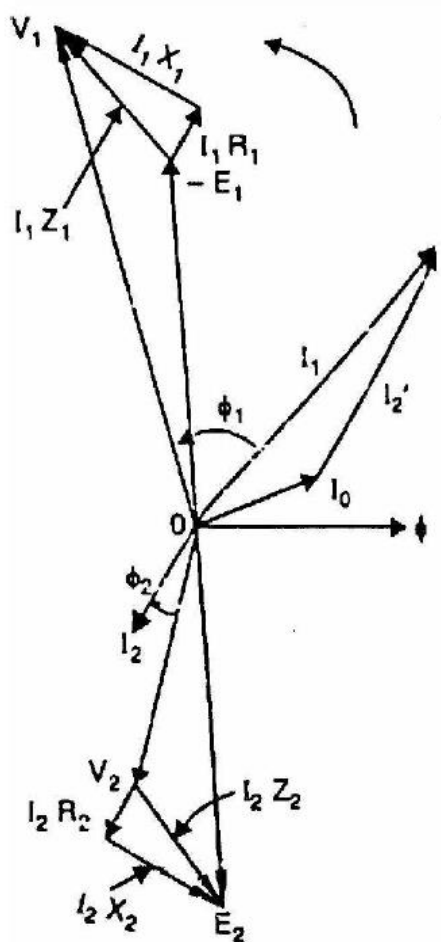
$$V_1 = E_1 + I_1 Z_1$$

$$V_2 = E_2 - I_2 (R_2 + j X_2) = E_2 - I_2 Z_2$$

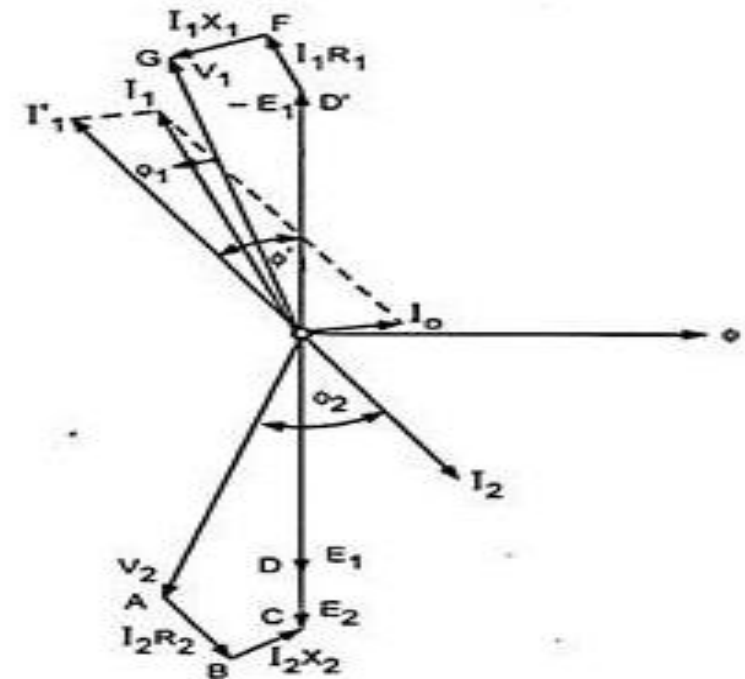


Vector Diagram

Inductive Load(Lagging Pf)



Capacitive Load(Leading Pf)



Transformer losses

The power losses in a transformer are of two types, namely;

1. Core or Iron losses
2. Copper losses

These losses appear in the form of heat and produce (i) an increase in temperature and (ii) a drop in efficiency.

1. Core or Iron losses (P_i)

These consist of hysteresis and eddy current losses and occur in the transformer core due to the alternating flux

$$\text{Hysteresis loss, } = k_h f B_m^{1.6} \text{ watts/m}^3$$

$$\text{Eddy current loss, } = k_e f^2 B_m^2 t^2 \text{ watts/m}^3$$

Both hysteresis and eddy current losses depend upon (i) maximum flux density B_m in the core and (ii) supply frequency f . Since transformers are connected to constant-frequency, constant voltage supply, both f and B_m are constant. Hence, core or iron losses are practically the same at all loads.

Iron or Core losses, $P_i = \text{Hysteresis loss} + \text{Eddy current loss}$
 $= \text{Constant losses}$

The hysteresis loss can be minimized by using steel of high silicon content whereas eddy current loss can be reduced by using core of thin laminations.

2. Copper losses

These losses occur in both the primary and secondary windings due to their ohmic resistance.

$$\begin{aligned} \text{Total Cu losses, } P_C &= I_1^2 R_1 + I_2^2 R_2 \\ &= I_1^2 R_{01} \text{ or } I_2^2 R_{02} \end{aligned}$$

Total losses in a transformer $= P_i + P_C$
 $= \text{Constant losses} + \text{Variable losses}$

Efficiency

F.L. Iron loss = P_i ...from open-circuit test

F.L. Cu loss = P_C ...from short-circuit test

Total F.L. losses = $P_i + P_C$

We can now find the full-load efficiency of the transformer at any p.f. without actually loading the transformer.

$$\text{F.L. efficiency, } \eta_{\text{F.L.}} = \frac{\text{Full - load VA} \times \text{p.f.}}{(\text{Full - load VA} \times \text{p.f.}) + P_i + P_C}$$

Also for any load equal to x full-load,

Corresponding total losses = $P_i + x^2 P_C$

$$\text{Corresponding } \eta_x = \frac{(x \text{ Full - load VA}) \times \text{p.f.}}{(x \text{ Full - load VA} \times \text{p.f.}) + P_i + x^2 P_C}$$

Note that iron loss remains the same at all loads.

Condition for maximum efficiency

Output power = $V_2 I_2 \cos \phi_2$

If R_{02} is the total resistance of the transformer referred to secondary,

Total Cu loss, $P_C = I_2^2 R_{02}$

Total losses = $P_i + P_C$

$$\begin{aligned} \therefore \text{Transformer } \eta &= \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{02}} \\ &= \frac{V_2 \cos \phi_2}{V_2 \cos \phi_2 + P_i / I_2 + I_2 R_{02}} \end{aligned}$$

$$\frac{d}{dI_2} (\text{denominator}) = 0$$

$$\text{or } \frac{d}{dI_2} (V_2 \cos \phi_2 + P_i / I_2 + I_2 R_{02}) = 0$$

$$0 - \frac{P_i}{I_2^2} + R_{02} = 0$$

$$P_i = I_2^2 R_{02}$$

Iron losses = Copper losses

Output KVA corresponding to maximum efficiency

Let P_C = Copper losses at full-load kVA

P_i = Iron losses

x = Fraction of full-load kVA at which efficiency is maximum

$$\text{Total Cu losses} = x^2 P_C$$

$$x^2 P_C = P_i \quad \dots \text{for maximum efficiency}$$

or
$$x = \sqrt{\frac{P_i}{P_C}} = \sqrt{\frac{\text{Iron loss}}{\text{F.L. Cu loss}}}$$

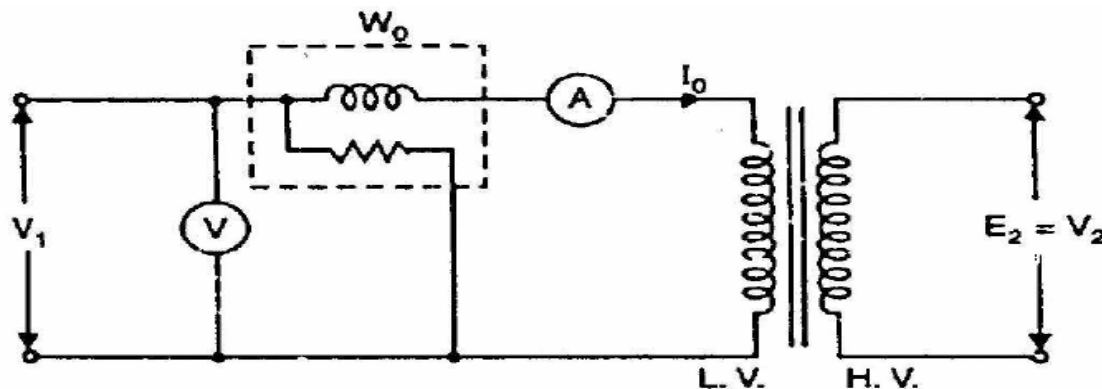
$$\text{Output kVA corresponding to maximum efficiency} = \text{Full - load kVA} \times \sqrt{\frac{\text{Iron loss}}{\text{F.L. Cu loss}}}$$

Transformer tests

- The circuit constants, efficiency and voltage regulation of a transformer can be determined by two simple tests
- (i) open-circuit test and (ii) short-circuit test.
- These tests are very convenient as they provide the required information without actually loading the transformer.
- Further, the power required to carry out these tests is very small as compared with full-load output of the transformer.

Open Circuit Test

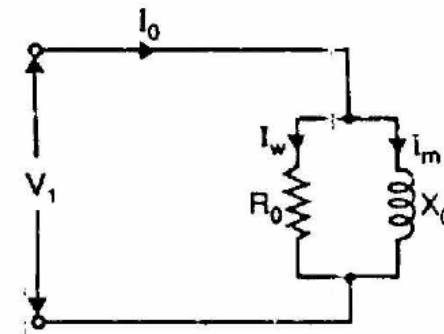
- This test is conducted to determine the iron losses (or core losses) and parameters R_0 and X_0 of the transformer.
- In this test, the rated voltage is applied to the low-voltage winding usually while the high voltage side is kept open.
- As the normal rated voltage is applied to the primary, therefore, normal iron losses will occur in the transformer core. Hence wattmeter will record the iron losses and small copper loss in the primary. Since no-load current I_0 is very small (usually 2-10 % of rated current). Cu losses in the primary under no-load condition are negligible as compared with iron losses.
- Hence, wattmeter reading W_0 practically gives the iron losses in the transformer



- The applied primary voltage V_1 is measured by the voltmeter, the no-load current I_0 by ammeter and no-load input power W_0 by wattmeter

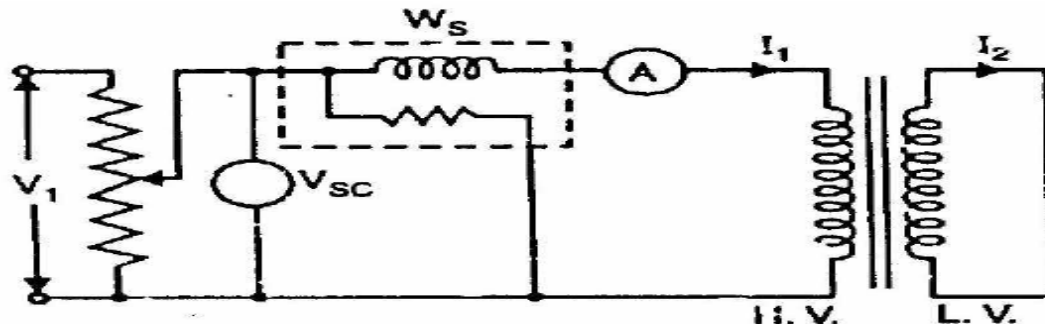
$$\begin{aligned} \text{Iron losses, } P_i &= \text{Wattmeter reading} = W_0 \\ \text{No load current} &= \text{Ammeter reading} = I_0 \\ \text{Applied voltage} &= \text{Voltmeter reading} = V_1 \\ \text{Input power, } W_0 &= V_1 I_0 \cos \phi_0 \end{aligned}$$

$$\begin{aligned} \cos \phi_0 &= \frac{W_0}{V_1 I_0} \\ I_w &= I_0 \cos \phi_0; \quad I_m = I_0 \sin \phi_0 \\ R_0 &= \frac{V_1}{I_w} \quad \text{and} \quad X_0 = \frac{V_1}{I_m} \end{aligned}$$



Short Circuit Test

- This test is conducted to determine R_{01} (or R_{02}), X_{01} (or X_{02}) and full-load copper losses of the transformer
- In this test, usually low voltage winding is short circuited by thick conductor and low voltage is applied to high voltage winding.
- The low input voltage is gradually raised till at voltage V_{sc} , full-load current I_{sc} flows in the winding.
- There is no output from the transformer under short-circuit conditions. Therefore, input power is all loss and this loss is almost entirely copper loss. It is because iron loss in the core is negligibly small since the voltage V_{sc} is very small. Hence, the wattmeter will practically register the full-load copper losses in the transformer windings

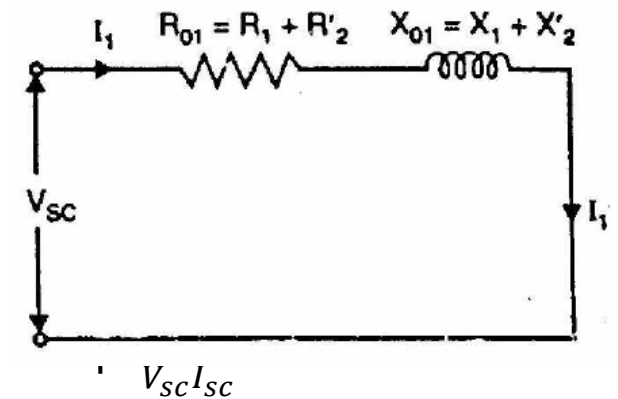


Full load Cu loss, P_C = Wattmeter reading = W_{sc}
 Applied voltage = Voltmeter reading = V_{sc}
 F.L. primary current = Ammeter reading = I_1

$$R_{sc} = \frac{W_{sc}}{I_{sc}^2}$$

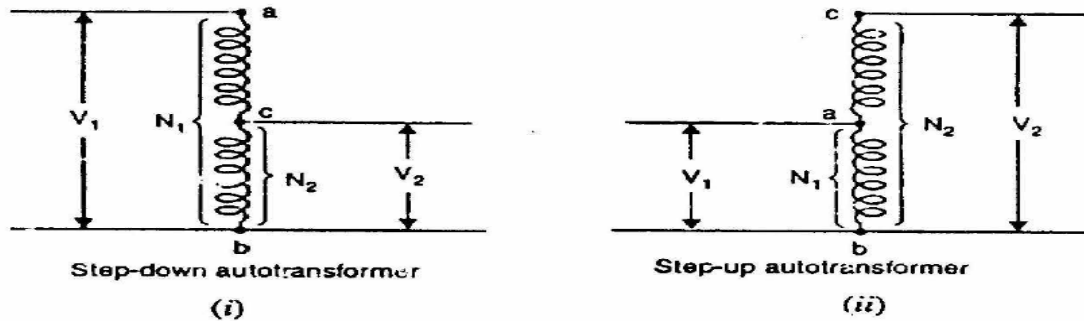
$$X_{sc} = \sqrt{Z_{sc}^2 - R_{sc}^2}$$

$$Z_{sc} = \frac{V_{sc}}{I_{sc}}$$



AUTOTRANSFORMER

- An autotransformer has a single winding on an iron core and a part of winding is common to both the primary and secondary circuits

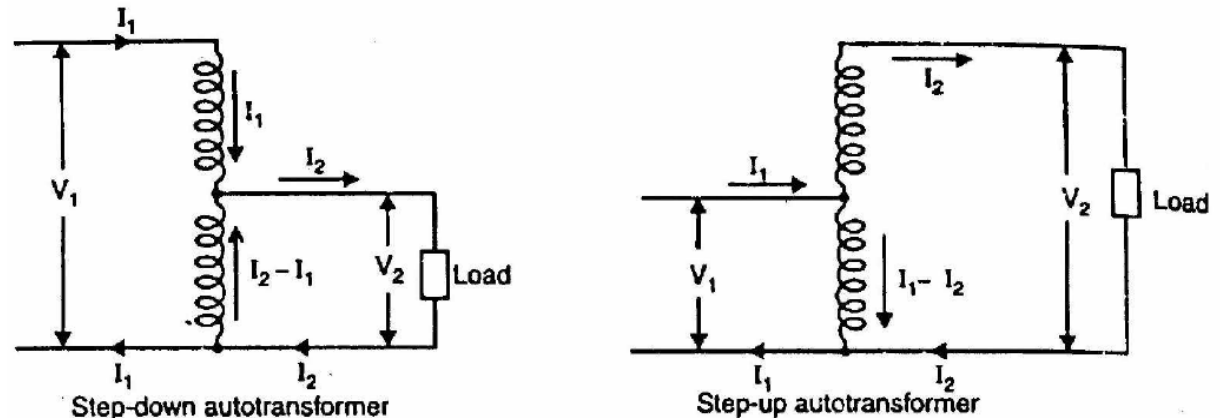


- The winding ab having N_1 turns is the primary winding and winding bc having N_2 turns is the secondary winding.
- The primary and secondary windings are connected electrically as well as magnetically. Therefore, **power from the primary is transferred to the secondary conductively as well as inductively**
- The voltage transformation ratio K of an ideal autotransformer is

$$K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

- I_1 is the input current and I_2 is the output or load current. Regardless of autotransformer connection (step-up or step-down), the current in the portion of the winding that is common to both the primary and the secondary is the difference between these currents (I_1 and I_2). The relative direction of the current through the common portion of the winding depends upon the connections of the autotransformer.

- For step-down autotransformer $I_2 > I_1$ (as for 2-winding transformer) so that $I_2 - I_1$ current flows through the common portion of the winding. For step-up autotransformer, $I_2 < I_1$. Therefore, $I_1 - I_2$ current flows in the common portion of the winding



Working of autotransformer

Consider step down transformer

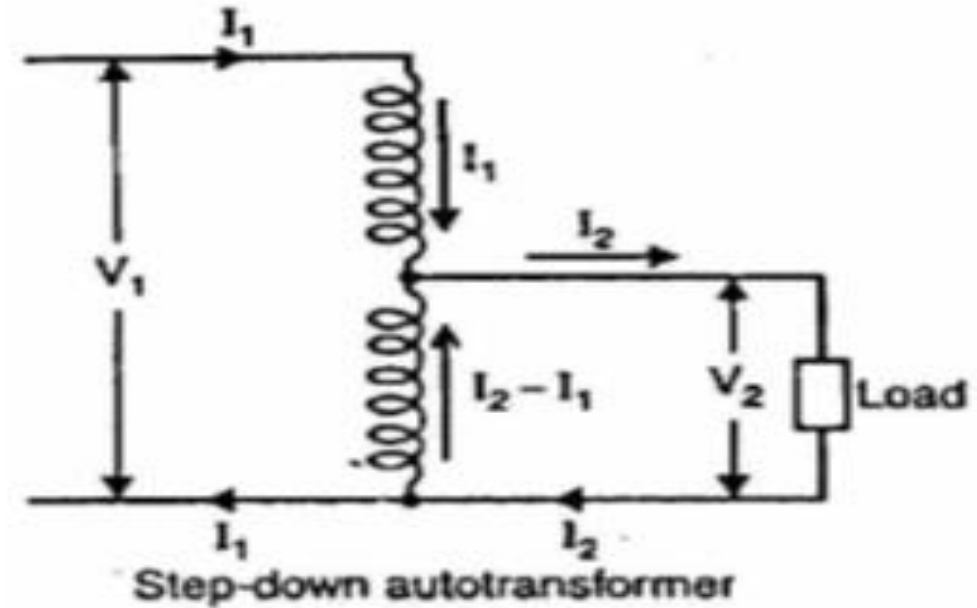
Since the primary and secondary windings of an autotransformer are connected magnetically as well as electrically, the power from primary is transferred to the secondary inductively (transformer action) as well as conductively (i.e., conducted directly from source to the load).

$$\text{Output apparent power} = V_2 I_2$$

$$\begin{aligned}\text{Apparent power transferred inductively} &= V_2(I_2 - I_1) = V_2(I_2 - K I_2) \\ &= V_2 I_2(1 - K) = V_1 I_1(1 - K)\end{aligned}$$

$$\therefore \text{Power transferred inductively} = \text{Input} \times (1 - K)$$

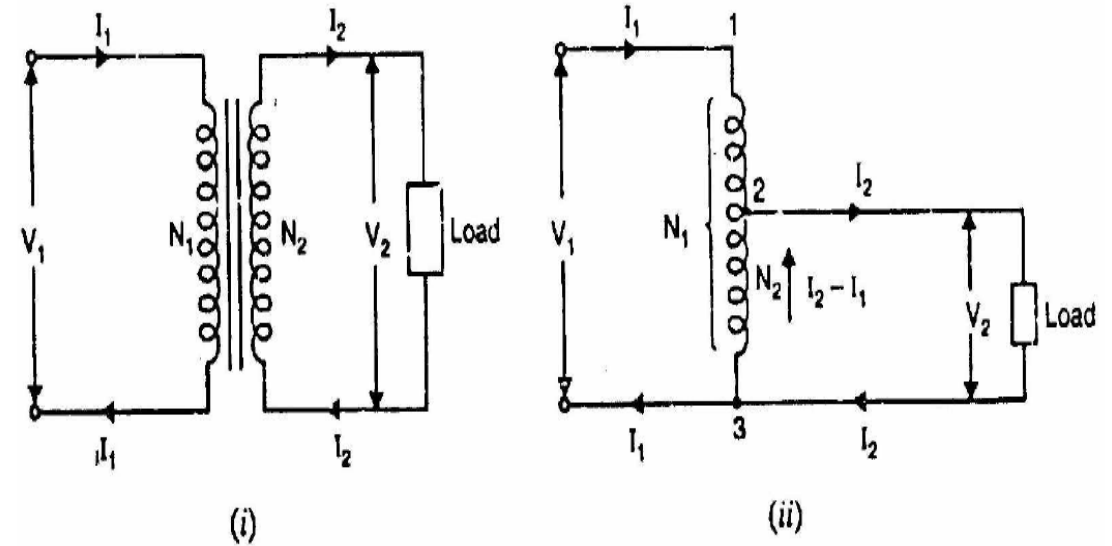
$$\begin{aligned}\therefore \text{Power transferred conductively} &= \text{Input} - \text{Input}(1 - K) \\ &= \text{Input} [1 - (1 - K)] \\ &= K \times \text{Input}\end{aligned}$$



Savings of copper in autotransformer

- Fig(i) shows an ordinary 2-winding transformer whereas Fig.(ii) shows an autotransformer having the same output and voltage transformation ratio K .
- For the same output and voltage transformation ratio $K(N_2/N_1)$, an autotransformer requires less copper than an ordinary 2-winding transformer.
- The length of copper required in a winding is proportional to the number of turns and the area of cross-section of the winding wire is proportional to the current rating. Therefore, the volume and hence weight of copper required in a winding is proportional to current $\times$ turns

Weight of Cu required in a winding $\propto$ current $\times$ turns



Winding transformer

$$\text{Weight of Cu required} \propto (I_1 N_1 + I_2 N_2)$$

Autotransformer

$$\text{Weight of Cu required in section 1-2} \propto I_1 (N_1 - N_2)$$

$$\text{Weight of Cu required in section 2-3} \propto (I_2 - I_1) N_2$$

$$\therefore \text{Total weight of Cu required} \propto I_1 (N_1 - N_2) + (I_2 - I_1) N_2$$

$$\begin{aligned}
\frac{\text{Weight of Cu in autotransformer}}{\text{Weight of Cu in ordinary transformer}} &= \frac{I_1(N_1 - N_2) + (I_2 - I_1)N_2}{I_1N_1 + I_2N_2} \\
&= \frac{N_1I_1 - N_2I_1 + N_2I_2 - N_2I_1}{N_1I_1 + N_2I_2} \\
&= \frac{N_1I_1 + N_2I_2 - 2N_2I_1}{N_1I_1 + N_2I_2} \\
&= 1 - \frac{2N_2I_1}{N_1I_1 + N_2I_2} \\
&= 1 - \frac{2N_2I_1}{2N_1I_1} \quad (\because N_2I_2 = N_1I_1) \\
&= 1 - \frac{N_2}{N_1} = 1 - K
\end{aligned}$$

$$\begin{aligned}
\therefore \text{Wt. of Cu in autotransformer (W}_a\text{)} \\
&= (1 - K) \times \text{Wt. in ordinary transformer (W}_o\text{)}
\end{aligned}$$

or $W_a = (1 - K) \times W_o$

$\therefore$ Saving in Cu = $W_o - W_a = W_o - (1 - K)W_o = K W_o$

or Saving in Cu = $K \times \text{Wt. of Cu in ordinary transformer}$

Advantages and disadvantages of autotransformer

Advantages

- (i) An autotransformer requires less Cu than a two-winding transformer of similar rating.
- (ii) An autotransformer operates at a higher efficiency than a two-winding transformer of similar rating.
- (iii) An autotransformer has better voltage regulation than a two-winding transformer of the same rating.
- (iv) An autotransformer has smaller size than a two-winding transformer of the same rating.
- (v) An autotransformer requires smaller exciting current than a two-winding transformer of the same rating.

Disadvantages:-

- Direct connection between primary and secondary
- Short circuiting problem is higher
- Not safe for stepping down a high voltage to low voltage.
- Insulating cost is high.

All Day Efficiency

- For instance, distribution transformers used for supplying lighting loads have their primaries energized all the 24 hours in a day but the secondaries supply little or no load during the major portion of the day.
- It means that a constant loss (i.e., iron loss) occurs during the whole day but copper loss occurs only when the transformer is loaded and would depend upon the magnitude of load.
- Consequently, the copper loss varies considerably during the day and the commercial efficiency of such transformers will vary from a low value (or even zero) to a high value when the load is high.
- The performance of such transformers is judged on the basis of energy consumption during the whole day (i.e., 24 hours). This is known as all-day or energy efficiency

The ratio of output in kWh to the input in kWh of a transformer over a 24-hour period is known as all-day efficiency i.e.,

$$\eta_{\text{all-day}} = \frac{\text{kWh output in 24 hours}}{\text{kWh input in 24 hours}}$$